

CS-402 Theroy Of Automata Update MCQS For Mid Term Solve By Vu Topper RM

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In FA, final state is represented by a _____ sign.

- A. -
- B. =
- C. *
- D. +**

If we have the regular expression $(a+b)^* (aaa + bbb) (a+b)^*$, then we can draw a TG for the provided RE with minimum _____ number of state(s).

- A. two
- B. zero
- C. one
- D. three**

Suppose we have the regular expression:

$aa(a+b+c)^*bb(a+b+c)^*cc$

Which of the following string will not be generated by the given RE?

- A. aabbcc
- B. aaaabbccbc
- C. aaabcbcbacc
- D. aaabbbbccc**

ok

If an FA has 3 states and 2 letters in the alphabet set, then it will have total _____ number of transitions.

- A. 7
- B. 4
- C. 6**
- D. 5

ok

Which one of the following string is a part of EQUAL language defined over $\Sigma=\{a,b\}$?

- A. aabbbaa
- B. aabbbbaa

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C. ababab

D. babab

Which one of the following word is not accepted by the given regular expression? $aa(a+b)^*b$

Aabbabb

Abbaab

Aaabab

aaaababb

Which of the following statement is NOT true about TG?

There may exist NULL path

There may be no final state

There exists exactly one path for certain string **ok**

There may exist more than one paths for certain string

Kleene's Theorem Part III expresses the relationship between _____.

A. RE and FA **ok**

B. FA and GTG

C. TG and RE

D. FA and TG

Which one of the following word is not accepted by the given regular expression? $(a+b)^*bba(a+b)^*$

Aaababab

Babbba

Abbbbbaa

bbabbbbaba

When even length language is expressed by an FA, then it will have minimum _____ states.

One

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Two

Four

Three

FA stands for _____.

A. Fixed Automaton

B. Finite Automaton

C. False Automaton

D. Functional Automaton

FA corresponding to an NFA can be built by introducing an empty state for a letter having

A. no transition at certain state

ok

B. one transition at certain state

C. two transitions at certain state

D. more than two transitions at certain state

In order to make NFA for the union of FA1 and FA2, the final state/ states of:

A. both FAs should be linked

B. both FAs should be left intact

C. FA1 have a transition to the final state of FA2

D. FA2 have a transition to the final state of FA1

Suppose we have FA3 which is equal to the union of FA1 and FA2. Now the initial state of the FA3 will be equal to:

A. Only initial state of FA1

B. Final state of FA2

C. Only initial state of FA2

D. Initial state of FA1 or FA2

ok

Introducing new final state in case of multiple final states is the step no. _____ of proving Kleene's theorem part II.

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A. 4 **ok**

B. 1

C. 2

D. 3

In proving Kleene Theorem II, circuits are reduced into:

A. asterisk

B. difference

C. both difference and asterisk

D. sum **ok**

Finite Automaton (FA) has:

A. Zero or more final states

B. Exactly one final state

C. Not more than two final states

D. Exactly two final states

The formal language is also known as _____.

A. Semantic language

B. Informal language

C. Syntactic language **ok**

D. Normal language

The language of all strings defined over alphabet set = {x, y} having triple x's or triple y's will have the minimum strings with length of:

A. 1

B. 3 **ok**

C. 4

D. 2

If an alphabet has "2" number of letters, then total number of strings of length "3" will be _____.

A. 5

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B. 9

C. 8

D. 6

GTG for the expression $(a+b)^*bb$ may have minimum number of states:

A. 3

B. 4

C. 2

D. 1

Consider the Regular Expression (RE) $a(a+b)^*$. Which one of the following is NOT accepted by the provided RE?

A. aa

B. aab

C. aba

ok

D. abb

Which of the following is NOT true about the term alphabet?

A. It is usually denoted by Greek letter sigma

B. It can be an empty set

C. Strings are generated by concatenating its elements

D. It is a finite set of symbols

Which of the following is free of non-determinism?

A. FA

ok

B. TG

C. NFA

D. NFA- $\wedge$

The language of all strings defined over alphabet set = $\{a, b\}$ containing 'bbb' will have the minimum string with length of:

A. 1

B. 2

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C.3 **ok**

D.4

The $aa(a+b)^*bb$ is the RE of the language defined over $\Sigma=\{a,b\}$. The language must _____.

A. have at least two ab

B. have at least one aa and one bb

C. have at least one abbb

D. have at least one ba

Reverse of string "YxwzYz" defined over $\Sigma = \{w,x,Y,z\}$ is _____.

A. zYzwxY **ok**

B. zYzwxY

C. zYwzxY

D. zYzwYx

If "r1" and "r2" are regular expressions, then which of the following is not a regular expression?

A. $r1 + r2$

B. $r1^*$

C. $r1 r2$

D. $r1 - r2$ **ok**

Which of the following string belongs to the language of the regular expression $(aa^*b)^*$?

A. baabab

B. aabaab

C. aaaaaa

D. abbbaa

FA of EVEN-EVEN language shows that it accepts the null string by declaring the _____ as a _____ as well.

A. Initial state, final state **ok**

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- B. Initial state, null state
- C. Final state, initial state
- D. Final state, null state

The language of all strings defined over alphabet set = {x, y} that ends with different letters will have the maximum length of:

- A. 1
- B. 2** ok
- C. infinite
- D. 3

In an FA, when there is no path from the initial state to final state, then that FA_____.

- A. accept all non empty strings
- B. does not accept any string** ok
- C. accept all strings
- D. accept null strings

If $\Sigma = \{a, b, c, d\}$. How many transitions will be there on each state of a finite automaton for any language defined over Σ ?

- A. 2
- B. 4** ok
- C. 1
- D. 3

Which of the following is the minimal number of states for a finite automaton accepting the language of all strings defined over any alphabet set?

- A. 3
- B. 4
- C. 2
- D. 1** ok

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How many states of a finite automaton will be final for accepting $L = \{\wedge, b, bb, bbb\}$?

A. 3

B. 1

C. 4 ok

D. 2

There is no compulsion that each state must have an outgoing edge for every input variable in:

A. Transition Graph

B. Transition Table

C. Both Finite Automata and Transition Graph

D. Finite Automata ok

In TG, there can be more than one _____.

A. start state only

B. null state only

C. start state and final state

D. final state only ok

Substrings as input letters can be specified on edges in:

A. NFA ok

B. PDA

C. FA

D. TG

If we have the regular expression $(a + b)^*$, then we can draw FA for the provided RE with minimum _____ number of state(s).

A. 2

B. 0

C. 1

D. 3

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Question No:1

(Marks:1)

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$\Sigma = \{a, Aa, Abb\}$, then string $aAaAbbAa$ has ___ length.

- A. One
- B. Two
- C. Three

D. Four

Page 4

Question No:2

(Marks:1)

Vu-Topper RM

Languages generated by kleene star are always___.

- A. Finite
- B. Infinite**
- C. Sometimes finite & sometimes infinite
- D. None of the these

Page 7

Question No:3

(Marks:1)

Vu-Topper RM

Let $S = \{aa, bb\}$, then S^* will have the _____ string.

- A. Λ**
- B. abba
- C. aabbbbaa
- D. bbaab

Page 7

ok

Question No:4

(Marks:1)

Vu-Topper RM

If $r1 = (aa + bb)$ and $r2 = (a + b)$ then the language $(aa + bb)^*$ will be generated by

- A. $(r1)(r2)$
- B. $(r1 + r2)$
- C. $(r2)^*$

D. $(r1)^*$

Page 10

Question No:5

(Marks:1)

Vu-Topper RM

If a language can be expressed through FA, then it can also be expressed through TG.

A. True

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B. False

Question No:6

(Marks:1)

Vu-Topper RM

If an alphabet has n number of letter, then number of strings of length m will be

- A. $n+m$
- B. $(n)(m)$
- C. m^n

D. n^m Page 6

Question No:7

(Marks:1)

Vu-Topper RM

In GTG, if a state has more than one incoming transitions from a state. Then all those incoming transitions can be reduced to one transition using sign

A. -

B. + Page 27

C. *

D. ()

Question No:8

(Marks:1)

Vu-Topper RM

Above given FA accepts _____ strings defined over $\Sigma = \{a, b\}$

A. All Page 15

B. Some

C. All but not null

D. None of these

Question No:9

(Marks:1)

Vu-Topper RM

One FA has 3 states and 2 letters in the alphabet. Then FA will have _____ number of transitions in the diagram

A. 4

B. 5

C. 7

D. 6 Page 14

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Question No:10 (Marks:1)

Vu-Topper RM

Every FA should be_____.

- A. Deterministic** Page 25 ok
B. Non- Deterministic
C. Deterministic & Non- Deterministic
D. None of these

Question No:11 (Marks:1)

Vu-Topper RM

Auto Meta mean

- A. Manual work
B. Automatic work Page 3
C. Both
D. None of these

Question No:12 (Marks:1)

Vu-Topper RM

NFA to FA will

- A. Equal** Page 43
B. Not equal
C. Not valid
D. None of given

Question No:13 (Marks:1)

Vu-Topper RM

The length of output string in case of__is one more than the length of corresponding input string.

- A. Finite Automaton** Page 55
B. TG
C. GTG
D. NFA

Question No:14 (Marks:1)

Vu-Topper RM

The__machine helps in building a machine that can perform the addition of binary numbers.

- A. Incrementing** Page 60

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- B. Complementing
- C. Decrementing
- D. None of the given

Question No:15

(Marks:1)

Vu-Topper RM

In proving Kleene Theorem II, if a state has two incoming transition edges labelled by RE from the same state, then replace all the edges with a single transition edge labelled by ----- of corresponding RE.

- A. Sum** **Page 27** **ok**
- B. Edge
 - C. FA
 - D. RE

Question No:16

(Marks:1)

Vu-Topper RM

Kleene Theorem III states that if the language can be expressed by RE then there exist ----- accepting the language.

- A. FA** **Page 32** **ok**
- B. DFA
 - C. NFA
 - D. None

Question No:17

(Marks:1)

Vu-Topper RM

If L_1 and L_2' are regular languages, $L_1 \cap (L_2' \cup L_1)'$ will be

- A. Regular** **Page 10**
- B. Ir-regular
 - C. Can't be decided
 - D. Another Language which is not listed here

Question No:18

(Marks:1)

Vu-Topper RM

A regular language can be:

- A. irregular
- B. infinite
- C. non-deterministic

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D. None of the given options

Question No:19

(Marks:1)

Vu-Topper RM

There _____ a language for which only FA can be built but not the RE.

A. is cannot be

B. is

C. may be ok

D. may not be

Question No:20

(Marks:1)

Vu-Topper RM

For every three regular expressions R, S, and T, the languages denoted by $R(S \cup T)$ and $(RS) \cup (RT)$ are the _____ .

A. Same

B. Different

C. $R(S \cup T)$ is Greater

D. None of the given options

Question No:21

(Marks:1)

Vu-Topper RM

In _____ there must be transition for all the letters of a string.

A. NFA

B. GTG

C. TG

D. FA

Question No:22

(Marks:1)

Vu-Topper RM

We cannot construct an NFA for the language of _____ defined over alphabet set $\{a,b\}$.

A. Even

B. odd

C. Palindromes ok

D. Integers

Question No:23

(Marks:1)

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Decomposing a string into its valid units is referred as:

- A. Decomposing
- B. Splitting
- C. Tokenizing**
- D. Dividing

Question No:24

(Marks:1)

Vu-Topper RM

Choose the correct word produced by RE $(a + b)^*$ (aa+bb).

- A. Abab
- B. Babab
- C. aaaa**
- D. Ab

Question No:25

(Marks:1)

Vu-Topper RM

Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum _____ number of states.

- A. 2
- B. 3
- C. more than 3**
- D. None of these

Question No:26

(Marks:1)

Vu-Topper RM

If R is a regular language and L is some language, and $L \cup R$ is a _____, then L must be a _____.

- A. Regular language**
- B. Finite Auto

Question No:27

(Marks:1)

Vu-Topper RM

The minimum length of the strings(except null string) of a language that starts and ends in different letters will be:

- A. 1**
- B. 2
- C. 3

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D. 4

Question No:28

(Marks:1)

Vu-Topper RM

Consider we have languages L_7 and L_6 . Which of the following represents their concatenation?

A. L_7+L_6 ok

B. L_7/L_6

C. L_6L_7

D. L_7L_6

Question No:29

(Marks:1)

Vu-Topper RM

Let FA1 has x number of states and FA2 has y number of states. Now FA_1+FA_2 can have maximum _____ number of states.

A. $x+y$

B. $x-y$

C. x/y

D. None ok

Question No:30

(Marks:1)

Vu-Topper RM

The language $\{a, ab, aba, bab\}$ is _____.

A. Irregular

B. Regular ok

C. Recursive

D. infinite

Question No:31

(Marks:1)

Vu-Topper RM

If we have a finite language and the number of states in the FA is n then the maximum number of letters in the each word of the language that will be accepted by the given FA will be:

A. N

B. $n-1$

C. $n+1$

D. 1

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Question No:32

(Marks:1)

Vu-Topper RM

Moore machine can have ----- final states.

A. 2

B. 4

C. 6

D. 8

Question No:33

(Marks:1)

Vu-Topper RM

There _____ be dead states in NFA.

A. may not

B. must

C. should not

D. will

Question No:34

(Marks:1)

Vu-Topper RM

Let L be the language of all strings, defined over $\Sigma = \{0,1\}$, ending in 10. Which of the following strings are distinguishable with respect to L with z being 0?

A. 010, 101

B. 111, 101

C. 001, 101

D. 111, 111

Question No:35

(Marks:1)

Vu-Topper RM

There _____ be a unique path for each valid string (called a word) in NFA.

A. May not

B. Must

C. Should not

D. Will

Question No:36

(Marks:1)

Vu-Topper RM

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If we have only one state, having no transition for input letters, then it is an example of:

A. RE

B. FA

C. TG

D. NFA ok

Question No:37

(Marks:1)

Vu-Topper RM

Strings x, y, z belongs to Σ^* such that $xz \in L$ but $yz \notin L$ where $L \subseteq \Sigma^*$ are:

A. Undetermined

B. Distinguishable

C. Indistinguishable

D. Both distinguishable and indistinguishable

Question No:38

(Marks:1)

Vu-Topper RM

A _____ with "n" states must accept at least one string of length greater than "n".

A. DFA ok

B. RE

C. Irregular language

D. Irrelevant language

Question No:39

(Marks:1)

Vu-Topper RM

In Moore machine, output is produced over the change of:

A. Transitions

B. Transitions and states

C. None of the mentioned

D. States

Question No:40

(Marks:1)

Vu-Topper RM

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 3 or more defined over

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$\Sigma = \{a, b\}$, with 'a' being the third letter from right?

- A. 10
- B. 12
- C. 14
- D. 16**

Question No:41

(Marks:1)

Vu-Topper RM

Every _____ is a _____ as well, but the converse may not be true.

- A. TG, FA
- B. GTG
- C. PDA
- D. FA, TG**

Page 19

ok

Question No:42

(Marks:1)

Vu-Topper RM

In the context of make NFA for the concatenation of FA1 and FA2 (FA1 accepting null string), which of the following option is correct?

- A. Final states in both FAs**
- B. Initial states in both FAs
- C. FA2 having initial state only
- D. FA2 having final state only

ok

Question No:43

(Marks:1)

Vu-Topper RM

In order to make NFA for the union of FA1 and FA2, the new initial state should be linked to:

- A. Initial states of both FAs**
- B. Initial and final states of FA1 and FA2 respectively
- C. Initial state of FA1 only
- D. Final and initial states of FA1 and FA2 respectively

ok

Question No:44

(Marks:1)

Vu-Topper RM

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 2 or more defined over

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$\Sigma = \{a, b\}$, with 'b' being the second letter from right?

- A. 9
- B. 6
- C. 7**
- D. 8

Question No:45

(Marks:1)

Vu-Topper RM

If we have an NFA having 3 states, and we convert that NFA to an FA.

The resultant FA will contains _____ states.

- A. 1
- B. 2** **ok**
- C. 3
- D. 4

Question No:46

(Marks:1)

Vu-Topper RM

Let FA3 be an FA corresponding to FA1FA2, then initial state of FA3 must correspond to the initial state of

- A. FA1 only
- B. FA2 only
- C. FA1 and FA2
- D. FA1 or FA2**

Question No:47

(Marks:1)

Vu-Topper RM

In which of the following machine, the length of output string is the same to that of input string?

- A. Mealy machine**
- B. Moore machine
- C. Finite automaton with output
- D. Non-deterministic finite automaton

Question No:48

(Marks:1)

Vu-Topper RM

Moore Machine is an application of:

- A. None of the mentioned

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B. Finite automata with output

C. Finite automata without input

Google

D. Non- Finite automata with output

Question No:49

(Marks:1)

Vu-Topper RM

In NFA having multiple transitions at certain state, FA can be built by introducing:

A. Empty state ok

B. Combination of states

C. Initial state

D. Final state

Question No:50

(Marks:1)

Vu-Topper RM

In Mealy machine the output depends on _____

A. Present state and Present input

B. Only present state

C. Nothing

D. Type of input

Question No:51

(Marks:1)

Vu-Topper RM

If L is a regular language, then $(L')^* \cup L$ will be:

A. L

B. C

C. P

D. F

Question No:52

(Marks:1)

Vu-Topper RM

A string will be accepted by an NFA if there exists _____ one successful path.

A. Atleast ok

B. Atmost

C. Maximum

D. None of the given options

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Question No:53

(Marks:1)

Vu-Topper RM

If A and B are regular languages, $!(A' \cup B')$ is:

- A. Non regular
- B. May be regular
- C. None of the mentioned
- D. Regular**

Question No:54

(Marks:1)

Vu-Topper RM

There is no question of accepting any language in:

- A. Moore machine**
- B. FA
- C. TG
- D. GTG

Question No:55

(Marks:1)

Vu-Topper RM

In _____ there must be transitions for all the alphabets over which a language is defined.

- A. FA ok**
- B. TG
- C. NFA
- D. GTG

Question No:56

(Marks:1)

Vu-Topper RM

Let FA3 be an FA corresponding to FA1FA2, then final state of FA3 must correspond to the final state of

- A. FA2 only**
- B. FA1 only
- C. FA1 or FA2
- D. FA1 and FA2

Question No:57

(Marks:1)

Vu-Topper RM

How many new states are introduced while developing NFA for the

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closure of an FA?

A. 2 **ok**

B. 4

C. 6

D. 8

Question No:58

(Marks:1)

Vu-Topper RM

Subtraction of binary numbers is possible through:

A. Both complementing and incrementing machine

B. Complementing machine

C. Incrementing machine

D. Converting machine

Question No:59

(Marks:1)

Vu-Topper RM

For a given Moore Machine, the input string is '101010', thus the output string would be of length:

A. Length of input string + 1

B. Length of input string – 1

C. Length of input string + 2

D. Length of input string -2

Question No:60

(Marks:1)

Vu-Topper RM

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

A. Mealy machine

B. Moore machine

C. Finite state machine

D. Deterministic finite state machine

Question No:61

(Marks:1)

Vu-Topper RM

If FA1 corresponds to $(a+b)^*$ then FA1 must accept _____ string/strings.

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- A. No
- B. Odd length
- C. Even length
- D. Every** ok

Question No:62 (Marks:1) **Vu-Topper RM**

Closure of an FA is the same as _____ of an FA with itself except that the initial state of the required FA is a final state as well.

- A. Sum
- B. Union
- C. Intersection
- D. Concatenation** ok

Question No:63 (Marks:1) **Vu-Topper RM**

Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

- 1. abaabaaabaa
 - 2. aaaabaaaa
 - 3. baaaaabaaaab
 - 4. baaaaabaa
- A. 1, 2 and 3
 - B. 2, 3 and 4
 - C. 1, 2 and 4** ok
 - D. 1, 3 and 4

Question No:64 (Marks:1) **Vu-Topper RM**

FA and _____ are same except that _____ has unique symbol for each transition.

- A. FA, TG
- B. NFA, TG
- C. NFA, FA** ok
- D. GTG, NFA

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Question No:65

(Marks:1)

Vu-Topper RM

How many states of a finite automaton will be final for accepting the only string 'abb', if $\Sigma = \{a, b\}$?

A. 1 **ok**

B. 2

C. 3

D. 4

Question No:66

(Marks:1)

Vu-Topper RM

Two machines are said to be equivalent if they print the output string when the input string is run on them.

A. Same, Same

B. Same, different

C. Different, same

D. Unique, different

Question No:67

(Marks:1)

Vu-Topper RM

Every NFA can be considered to be a ----- as well, but the converse may not be true.

A. TG **ok**

B. FA

C. GTG

D. PDA

Question No:68

(Marks:1)

Vu-Topper RM

In which of the following machine, the length of output string is 1 more than that of input string?

A. Mealy machine

B. Non-deterministic finite automaton

C. Finite automaton with output

D. Moore machine

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Question No:69

(Marks:1)

Vu-Topper RM

If $S = \{aa, bb\}$ then S^* will not contain _____.

A. abbbab

B. bbba

C. bbbbab

D. ababbb

Question No:70

(Marks:1)

Vu-Topper RM

Which of the following machine has only one initial state and no final state?

A. Moore machine

B. Finite state machine

C. Deterministic finite state machine

Question No:71

(Marks:1)

Vu-Topper RM

Which of the following diagram is very rigid in order to express any language?

A. TG

B. NFA

C. GTG

D. FA

ok

Question No:72

(Marks:1)

Vu-Topper RM

If $S = \{a\}$, then S^+ will be _____.

A. $\{a, aaa, aaaa, aaaaa, \dots\}$

B. $\{a, aa, aaa, aaaa, \dots\}$

ok

C. $\{a, aaa, aaaaa, aaaaaaa, \dots\}$

D. $\{aa, aaaa, aaaaaa, aaaaaaaa, \dots\}$

Question No:73

(Marks:1)

Vu-Topper RM

Let L be the language of all strings. defined over $\Sigma = \{0,1\}$. ending in 111. Melay machine can have _____ final states.

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A. Zero

B. One

C. More than one but finite

D. More than one but infinite

Question No:74

(Marks:1)

Vu-Topper RM

Let's we have two regular expressions $R1=(xx+yy)$ and $R2=(x+y)$. Which one of the following is the correct regular expression for the Union of $R1$ and $R2$?

A. $(xx+yy)(x+y)$

B. $(xx+yy)+(x+y)^*$

C. $(xx+yy)+(x+y)$

D. $((xx+yy)+(x+y))^*$

Question No:75

(Marks:1)

Vu-Topper RM

The state where there is no way to leave after entry, is called

_____.

A. Davey John locker

B. initial state

C. final state

D. non-final state

Question No:76

(Marks:1)

Vu-Topper RM

Which one of the following word is not accepted by the given regular expression?

A. aaabab

B. aaaababb

C. abbaab

D. aabbabb

Question No:77

(Marks:1)

Vu-Topper RM

According to theory of automata there are _____ types of languages.

A. One

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- B. Two
C. Three
D. Four

Question No:78 (Marks:1) Vu-Topper RM
Question No:79 (Marks:1) Vu-Topper RM

Regular languages are closed under the following operations.

- A. Union only
B. Concatenation, Closure only
C. Union, Concatenation and Closure ok
D. Regular languages are not closed under any operation

Question No:80 (Marks:1) Vu-Topper RM
There can be more than _____ FA for a certain language but for _____ FA there is only one language associated with it.

- A. one, one** ok
B. one, two
C. two, three
D. two, one

Question No:81 (Marks:1) Vu-Topper RM
There is one compulsion that each state must have an on outgoing edge forevery input variable in:

- A. Finite Automata
B. Transition Graph
C. Both Finite Automata and Transition Graph
D. Transition Table

Question No:82 (Marks:1) Vu-Topper RM
FA is also called

- A. TG
B. GTG
C. NFA

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D. DFA **ok**

Question No:83 **(Marks:1)** **Vu-Topper RM**

If r_1 and r_2 are regular expressions then $(r_1 * r_2)$ is _____ .

- A. FA
- B. TG
- C. GTG

D. RE **ok**

Question No:84 **(Marks:1)** **Vu-Topper RM**

Keep in view the language of all strings ending with 'a' defined over $\Sigma = \{a, b, c, d\}$. For which input letter, we will take a loop on the final state of its transition diagram?

- A. A** **ok**
- B. b
- C. c
- D. d

Question No:85 **(Marks:1)** **Vu-Topper RM**

Which of the following statements is true about NFA with Null String?

- A. Infinite states
- B. Infinite set of letters
- C. Infinite set of transitions

D. Transition of null string is allowed at any stage **ok**

Question No:86 **(Marks:1)** **Vu-Topper RM**

Introducing new start state in case of multiple start states is the step no. _____ of proving Kleene's theorem part II.

- A. 1** **ok**
- B. 2
- C. 3
- D. 4

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Question No:87

(Marks:1)

Vu-Topper RM

Which of the following diagrams expresses languages more simply?

- A. FA
- B. NFA
- C. TG
- D. GTG**

Question No:88

(Marks:1)

Vu-Topper RM

The language of all strings defined over alphabet set = {a, b} that does not end with 'a' actually ends with:

- A. b
- B. b and $\wedge$ ok**
- C. $\wedge$
- D. $\wedge$ and a

Question No:89

(Marks:1)

Vu-Topper RM

In NFA having no transition at certain state, FA can be built by introducing:

- A. Empty state ok**
- B. Combination of states
- C. Initial state
- D. Final state

Question No:90

(Marks:1)

Vu-Topper RM

Formal is also known as

- A. Syntactic language**
- B. Semantic language
- C. Informal language
- D. None of these

Question No:91

(Marks:1)

Vu-Topper RM

There may be more than one transition for a certain letter on a state in:

- A. Finite automata

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B. Non-Deterministic Finite Automata

ok

C. Transition Table

D. Moore Machine

Question No:92

(Marks:1)

Vu-Topper RM

FA of EVEN language shows null string when

A. Initial state is final as well

B. EVEN does not accept null

C. One state is declared null

D. None of the these

Question No:93

(Marks:1)

Vu-Topper RM

Which of the following statement is true about GTG?

A. Transitions are based on input letters

B. Transitions are based on specified substrings

ok

C. Transitions are based on regular expressions

D. Transitions are based on alphabet set

Question No:94

(Marks:1)

Vu-Topper RM

In GTG, there can be more than one:

A. Start state

B. Final state

C. Start state and final state

ok

D. Null state

Question No:95

(Marks:1)

Vu-Topper RM

GTG for the expression $(aa+aba)^*$ may have minimum number of states:

A. 1

ok

B. 2

C. 3

D. 4

Question No:96

(Marks:1)

Vu-Topper RM

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In regular expressions, the operator '*' stands for

A. Concatenation ok

B. Iteration

C. Selection

D. Add

Question No:97

(Marks:1)

Vu-Topper RM

If r_1 is a regular expression then $(r_1)^*$ is _____.

A. A generalized transition graph

B. A non-deterministic finite automaton

C. A finite automaton

D. Also, a regular expression ok

Question No:98

(Marks:1)

Vu-Topper RM

Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

A. 1

B. 2

C. 3

D. 4 ok

Question No:99

(Marks:1)

Vu-Topper RM

Kleene's theorem states that

A. All representations of a regular language are equivalent. ok

B. Finite Automata are less powerful than Pushdown Automata.

C. All representations of a context free language are equivalent.

D. All representations of a recursive language are equivalent

Question No:100

(Marks:1)

Vu-Topper RM

Melay machine to increase the output string in magnitude by 1 is called:

A. Complementing machine

B. Incrementing machine

C. Decrementing machine

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D. Converting machine

Question No:101

(Marks:1)

Vu-Topper RM

Kleene's Theorem Part I expresses the relationship between_____.

A. FA and TG

B. TG and RE

C. RE and FA

D. FA and RE

Question No:102

(Marks:1)

Vu-Topper RM

Suppose we have FA3 (which is equal to FA1 + FA2), then the final state of FA3 will be declared final if:

A. It corresponds to final states of both FA1 and FA2

B. It corresponds to final states of FA1 only

C. It corresponds to final states of FA2 only

D. It corresponds to any of the final states in FA1 or FA2 ok

Question No:103

(Marks:1)

Vu-Topper RM

Null strings can be specified on edges in:

A. Finite Automata

B. Non-Deterministic Finite Automata

C. Transition Graph

D. Melay Machine

Question No:104

(Marks:1)

Vu-Topper RM

What is false about the PALINDROME LANGUAGE?

A. Every word is reverse of itself.

B. It is an infinite language.

C. FA can be build for it.

D. None of the given option ok

Question No:105

(Marks:1)

Vu-Topper RM

While finding RE corresponding to TG, If TG has more than one startstate

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then

- A. Introduce the new start state** ok
- B. Eliminate the old start state
- C. Replace the old start state with final state
- D. Replace the old final state with new start state

Question No:106

(Marks:1)

Vu-Topper RM

All possible combinations of strings of a language including null string is referred as:

- A. Concatenation of a language with itself
- B. Kleene star closure of a language** ok
- C. Multiplication of language with itself
- D. Addition of a language with itself

Question No:107

(Marks:1)

Vu-Topper RM

$n!$ will be equal to:

- A. $n*n$
- B. $n*(-n)!$
- C. $n*(n-1)$
- D. $n*(n-1)!$** ok

Question No:108

(Marks:1)

Vu-Topper RM

While finding RE corresponding to a TG, we connect the new start state with the old start state by _____ transition.

- A. a
- B. b
- C. Null** ok
- D. RE

Question No:109

(Marks:1)

Vu-Topper RM

In proving Kleene Theorem II, if three states are connected then middle state is removed by connecting first and third state and writing corresponding RE in:

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A. Sum

B. Concatenation ok

C. Difference

D. Asterisk

Question No:110

(Marks:1)

Vu-Topper RM

In___ there must be transition for all the letters of a string.

A. NFA

B. GTG

C. TG

D. FA

Question No:111

(Marks:1)

Vu-Topper RM

There is no question accepting any language in:

A. FA

B. TG

C. GTG

D. Moore machine

Question No:112

(Marks:1)

Vu-Topper RM

The FA can be drawn for the regular expression $(a+b)^*$ with minimum state(s).

A. 1

B. 2

C. 3

D. 4

Question No:113

(Marks:1)

Vu-Topper RM

Which of the following does not contribute while finding out the length of strings?

A. ^ ok

B. a

C. b

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D. $a+b$

Question No:114

(Marks:1)

Vu-Topper RM

The language of all strings defined over alphabet set = $\{x, y\}$ that ends with same letters will have the maximum length of:

A. 1

B. 2

C. 3

D. Infinite ok

Question No:115

(Marks:1)

Vu-Topper RM

Considering FA1 and FA2 states each. Now FA1+FA2 can have maximum_number of states.

A. 2

B. 3

C. More than 3

D. None of the given option

Question No:116

(Marks:1)

Vu-Topper RM

Which one of the following is the RE for the language defined over $\Sigma = \{a, b\}$ having all the words starting with a?

A. $(a + b)^*$

B. $aa(a + b)^+$

C. $a(a + b)^*$ ok

D. $a^*(a + b)$

Question No:117

(Marks:1)

Vu-Topper RM

An _____ can be considered to be an intermediate structure between Finite automaton and Transition Graph.

A. RE

B. GTG

C. NFA ok

D. None of the given options

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Question No:118

(Marks:1)

Vu-Topper RM

Suppose a language L_1 has 2 states and L_2 has 2 states. If we have a machine M that accepts $L_1 \cap L_2$. Then, the total number of states in M is equal to _____.

A. 2

B. 4

C. 6

D. 8

Question No:119

(Marks:1)

Vu-Topper RM

FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having

A. No transition at certain state **ok**

B. One transition at certain state

C. Two transitions at certain state

D. More than two transitions at certain state

Question No:120

(Marks:1)

Vu-Topper RM

Automata is the plural of _____.

A. Automate

B. Automaton

C. Automation

D. Automatic

Question No:121

(Marks:1)

Vu-Topper RM

In NFA having no transition at certain. FA can be built by introducing:

A. Empty state

B. Combination of states

C. Initial state

D. Final state

Question No:122

(Marks:1)

Vu-Topper RM

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If $S = \{ x \}$, then S^* will be _____.

A. $\{^,x,xxx,xxxx,xxxxx,\dots\}$

B. $\{^,x,xx,xxx,xxxx,\dots\}$ ok

C. $\{^,x,xxx,xxxxx,xxxxxxxx,\dots\}$

D. $\{^,xx,xxxx,xxxxxx,xxxxxxxx,\dots\}$

Question No:123

(Marks:1)

Vu-Topper RM

In TG, the string is supposed to be _____ if there is no path for a string from initial to final state.

A. Accept null string

B. Accept all strings

C. Accept all non-empty strings

D. Does not accept any string

Question No:124

(Marks:1)

Vu-Topper RM

In Moore machine, if the length of input string is 9, then the length of output string will be:

A. 7

B. 8

C. 9

D. 10

Question No:125

(Marks:1)

Vu-Topper RM

When ODD language is expressed by an FA, then it will have minimum _____ states.

A. One

B. Two

C. Three

D. Four

Question No:126

(Marks:1)

Vu-Topper RM

$[(a + b)(a + b)]^*$, given RE cannot generate the string _____.

A. abbaabab

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B. abbbbaa

C. bbbbbb

ok

D. abbbaaaaa

Question No:127

(Marks:1)

Vu-Topper RM

The recursive method for defining a language has _____ steps.

A. One

B. Two

C. Three

D. Four

Question No:128

(Marks:1)

Vu-Topper RM

Consider the following RE:

$a(a + b)b^*$

All of the following words are accepted except_____.

A. aab

B. abb

C. aa

D. aba

Question No:129

(Marks:1)

Vu-Topper RM

For every three regular expressions R, S, T, the languages denoted by $R(S \cup T)$ and $(RS) \cup (RT)$ are the_____.

A. Same

B. Different

C. $R(S \cup T)$ is greater

D. None of the given options

Question No:130

(Marks:1)

Vu-Topper RM

Alphabet $S = \{a, bc, cc\}$ has _____ number of letters.

A. One

B. Two

C. Three

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D. Four

Question No:131

(Marks:1)

Vu-Topper RM

Two FAs are said to be equivalent, if they _____.

A. Accept null string

B. Accept same language **ok**

C. Accept different language

D. None of the given options

Question No:132

(Marks:1)

Vu-Topper RM

----- can also help in proving Kleene Theorem III.

A. NFA **ok**

B. PDA

C. Moore machine

D. Melay machine

Question No:133

(Marks:1)

Vu-Topper RM

Kleene's Theorem Part II expresses the relationship between

_____.

A. FA and TG

B. TG and RE **ok**

C. RE and FA

D. FA and RE

Question No:134

(Marks:1)

Vu-Topper RM

If two RE's generate same language then these RE's are called

_____.

A. Same RE

B. Equal RE

C. Similar RE

D. Equivalent RE

Question No:135

(Marks:1)

Vu-Topper RM

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Every FA should be__.

- A. Deterministic**
- B. Non-deterministic
- C. Deterministic and non-deterministic
- D. Not depends on language

Question No:136

(Marks:1)

Vu-Topper RM

What statement is true?

- A. A letter is always a combination of symbols**
- B. A letter may consist of one symbol
- C. There is no difference between symbol and letter
- D. Letters and symbols are the same thing

Question No:137

(Marks:1)

Vu-Topper RM

If $\Sigma = \{ab, bb\}$, then Σ^* will not contain

- A. abbbab
- B. bbba**
- C. bbbbab
- D. ababbb

Question No:138

(Marks:1)

Vu-Topper RM

Choose the correct word produced by RE $(a + b)^*ab$.

- A. abb
- B. abab**
- C. bbbb
- D. aaaa

Question No:139

(Marks:1)

Vu-Topper RM

According to 1st part of the Kleene's theorem, If a language can be accepted by an FA then it can be accepted by a__as well

- A. FA
- B. CFG
- C. GTG

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D. TG

Question No:140

(Marks:1)

Vu-Topper RM

“One language can be expressed by ___GTG”.

- A. Only one
- B. Only two
- C. **More than one**

Question No:141

(Marks:1)

Vu-Topper RM

If a TG has more than one start states, then we can make a single startstate by introducing a new state and connecting it with all the previously existing start states by using.

- A. Any infinite string
- B. Single letter string
- C. **Null string**
- D. Any finite string

Question No:142

(Marks:1)

Vu-Topper RM

If in a NFA, Λ is allowed to be a label of an edge then that NFA is called ____.

- A. TG
- B. RE
- C. **NFA with null string**
- D. RE

Question No:143

(Marks:1)

Vu-Topper RM

If we want to make a Moore machine equivalent to mealy machine then

- A. **We should ignore the extra character printed by the Moore machine.**
- B. We should ignore the extra character printed by the Mealy machine.
- C. We will make the initial state as a no carry state.
- D. We should not ignore the extra character printed by the Moore machine.

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Question No:144

(Marks:1)

Vu-Topper RM

Two machines are said to be equivalent if they print the_output string when same input string is run no them.

A. Same

B. Different

C. Inverse

D. Null

Question No:145

(Marks:1)

Vu-Topper RM

The length of output in case of___is one more than the length of corresponding input string

A. Moore machine

B. Mealy machine

C. Incremental machine

D. Adding machine

Question No:146

(Marks:1)

Vu-Topper RM

A is not a valid transition in

A. TG

B. GTG

C. NFA

D. RE

Question No:147

(Marks:1)

Vu-Topper RM

Dead states are also called

A. John Davey Lockers

B. Davey John Lockers

C. Mutex Lockers

D. Semaphores

Question No:148

(Marks:1)

Vu-Topper RM

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Language of all strings whose length is odd and number of y's is even defined over alphabet set $\Sigma = \{x, y\}$. _____ will be accepted by the given language.

A. xxyxyxyyyx

B. xxyxyxyyyxy

ok

C. xxyxyxyyyxx

D. xxyxyxyyy

Question No:149

(Marks:1)

Vu-Topper RM

If an effectively solvable problem has answer in Yes or NO. then the solution is called

A. Infinite problem

B. Decision procedure

C. Finite solution

D. Optimal procedure

Question No:150

(Marks:1)

Vu-Topper RM

If the intersection of two regular languages is regular then the complement of the intersection of these two languages is

A. Regular

B. Irregular

C. Irregular but finite

D. Irregular but infinite

Question No:151

(Marks:1)

Vu-Topper RM

If R is regular language and Q is any language (regular/non-regular). Then Pref(in____) is regular.

A. Q, Q

B. Q, R

C. R, Q

D. R, R

Question No:152

(Marks:1)

Vu-Topper RM

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The strings or words which do not belong to a language are called of that language

- A. Intersection
- B. Union
- C. Complement**
- D. Quotient

Question No:153

(Marks:1)

Vu-Topper RM

Prime is a _language.

- A. Finite
- B. Both context free and regular
- C. Regular
- D. Non-regular**

Question No:154

(Marks:1)

Vu-Topper RM

Finite Automaton (FA) must have ____ number of states while a language has __ words.

- A. Infinite, finite
- B. Finite, finite
- C. Finite, infinite**
- D. Infinite, infinite

Question No:155

(Marks:1)

Vu-Topper RM

The language "PRIME" is an example of ____ language.

- A. Regular but finite
- B. Regular
- C. Non regular but finite
- D. Non regular**

Page 75

Question No:156

(Marks:1)

Vu-Topper RM

If L1 and L2 are regular languages then which statement is NOT true?

- A. $L1 + L2$ is always regular
- B. $L1 L2$ is always regular

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C. L1/L2 is always regular

D. L1* is always regular

Question No:157

(Marks:1)

Vu-Topper RM

If a language is regular it must generate _____ number of distinct classes.

A. Finite

B. Infinite

C. Two

D. three

Question No:158

(Marks:1)

Vu-Topper RM

The operators like (* . +) in the parse tree are considered as

A. Terminals

B. Non-terminals

C. Productions

D. Intermediates

Question No:159

(Marks:1)

Vu-Topper RM

Set of all palindromes over {a,b} is:

A. Regular

B. Regular and finite

C. Regular and infinite

D. Non-regular

Question No:160

(Marks:1)

Vu-Topper RM

Which one of the following languages is a non-regular language?

A. Even-even

B. Containing double a

C. Start and end with same letter

D. Palindrome

Question No:161

(Marks:1)

Vu-Topper RM

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The language of all strings partition Σ^* into_class(es).

A. One

B. Two

C. Three

D. Four

Question No:162

(Marks:1)

Vu-Topper RM

The language of all strings not beginning with 'b' partitions Σ^* into distinct classes.

A. Two

B. Three

C. Four

D. Five

Question No:163

(Marks:1)

Vu-Topper RM

The values of input (say a & b) do not remain same in one cycle due to

A. NAND gate

B. Clock pulse

C. OR gate

D. NOT gate

Question No:164

(Marks:1)

Vu-Topper RM

In a CFG, the non-terminals are denoted by

A. Small letters

B. Numbers

C. Capital letters

D. Small letters and numbers

Question No:165

(Marks:1)

Vu-Topper RM

$a^* + b^* = (a + b)^*$ this expression is

A. True

B. False

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Question No:166

(Marks:1)

Vu-Topper RM

Length of EVEN-EVEN language is always_____.

- A. Even** **ok**
- B. Odd
- C. Sometimes even & sometimes odd
- D. Such language doesn't exist

Question No:167

(Marks:1)

Vu-Topper RM

While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by

- A. a
- B. b
- C. null
- D. none of the given options**

Question No:168

(Marks:1)

Vu-Topper RM

Given S, Kleene star closure is denoted by_____.

- A. S*** **ok**
- B. S+
- C. S-
- D. None of these

Question No:169

(Marks:1)

Vu-Topper RM

Which of the following steps replaces multiple incoming transition edges with a single one in proving Kleene's theorem part ||?

- A. 1
- B. 2
- C. 3
- D. 4**

Question No:170

(Marks:1)

Vu-Topper RM

If $r_1 = (aa + bb)$ and $r_2 = (a + b)$ then the language $(aa + bb)(a + b)$ will be generated by _____.

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A. (r1)(r2)

ok

B. (r1 + r2)

C. (r2)(r1)

D. (r1)*

Question No:171

(Marks:1)

Vu-Topper RM

The language having even number of a's and even number of b's defined over $S = \{a, b\}$ is called _____.

A. EVEN-EVEN

ok

B. ODD-ODD

C. PALINDROME

D. FACTORIAL

Question No:172

(Marks:1)

Vu-Topper RM

If L_1' and L_2' are regular languages. Then L_1, L_2 will be

A. Regular

B. Non regular

C. May be regular

D. None of the mentioned

Question No:173

(Marks:1)

Vu-Topper RM

If FA1 corresponding to $(a+b)^*$ then FA1 must accept string/strings

A. No

B. Odd length

C. Even length

D. Every

Question No:174

(Marks:1)

Vu-Topper RM

In FA, initial state can be represented by:

A. Drawing an arrow head before that state

B. Drawing a circle in that state

C. leave the state empty

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D. Drawing '+' sign in that state

Question No:175

(Marks:1)

Vu-Topper RM

An FA is a collection of:

A. Finite states, finite transition and finite input letters ok

B. Infinite states, infinite transition and infinite input letters

C. Only finite states and finite transitions

D. Only infinite states and infinite transitions

Question No:176

(Marks:1)

Vu-Topper RM

NFA with null string has ----- initial state(s).

A. One ok

B. Two

C. Four

D. Three

Question No:177

(Marks:1)

Vu-Topper RM

The difference between number of states with regular expression $(a + b)$ and $(a + b)^*$ is:

A. 0

B. 1

C. 2

D. 3

Question No:178

(Marks:1)

Vu-Topper RM

A transition graph is converted into a(n) _____ in order to obtain regular expression.

A. FA

B. GTG

C. NFA

D. NFA ok

Question No:179

(Marks:1)

Vu-Topper RM

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Consider the languages $L1 = \{a\}$ and $L2 = \{a\}$. Which one of the following represents $L1 L2^* \cup L1^*$

A. $\wedge$

B. a^*

C. All of the mentioned

D. None of the mentioned

Question No:180

(Marks:1)

Vu-Topper RM

If $S = \{a, b\}$ then which of the following RE will generate all possible strings?

A. $a^* + b^*$

B. $(ab)^*$

C. $(a + b)^*$

D. $(ab + ba)^*$

Question No:181

(Marks:1)

Vu-Topper RM

In drawing FA3 (which is equal to $FA1 + FA2$), a state will be declared final if

A. It corresponds to final states of both FA1 and FA2

B. It corresponds to final states of FA1

C. It corresponds to final states of FA2

D. It corresponds to any of the final states in FA1 or FA2

Question No:182

(Marks:1)

Vu-Topper RM

Let $S = \{a, bb, bab, baabb\}$ be a set of strings, which one of the following will not be included in S^* ?

A. baba

B. baabbabb

C. bbbaaabb

ok

D. bbbaabaabb

Question No:183

(Marks:1)

Vu-Topper RM

The length of string "AbBAbcd" defined over $\Sigma = \{A, b, B, c, d\}$ is

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ok

D. Four

Question No:184

(Marks:1)

Vu-Topper RM

In case of finite automaton there _____ be a transition on each _____ for every letter of the alphabet set.

A. Must, state

B. Maybe, state

C. Often, edge

D. Must, edge

Question No:185

(Marks:1)

Vu-Topper RM

Which one of the following word is not accepted by the given regular expression? $(a+b)^*(aaa+bbb)(a+b)^*$

A. Ababaaaab

B. Bababbbba

C. Baabaabba

ok

D. Abbaaabbba

Question No:186

(Marks:1)

Vu-Topper RM

1 Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:

A. FA1

B. FA2

C.FA2-FA1

D. (FA2)

Question No:187

(Marks:1)

Vu-Topper RM

Edges are expressed with a regular expression in:

A. GTG

Page 23

ok

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- B. FA
- C. NFA
- D. TG

Question No:188

(Marks:1)

Vu-Topper RM

NFA corresponding to union of FAs is built by introducing a new start state and connect it to the states originally connected to the old start state with the ----- transitions as the old start state:

- A. Same** **ok**
- B. Union of
- C. Different
- D. Concatenated

Question No:189

(Marks:1)

Vu-Topper RM

----- state is not important in Moore machine.

Final
Start

Question No:190

(Marks:1)

Vu-Topper RM

If we subtract a binary number 1010 from the binary number 1101(ignore the overflow), then the result will be:

1100
0011

Question No:191

(Marks:1)

Vu-Topper RM

In concatenation, we include the initial state of FA2 automatically after the final state of FA1 because of:

- A. We need just two initial states
- B. We need just one initial state** **ok**
- C. Some part of the string may be accepted by FA2
- D. The strings of FA2 are accepted first before the strings of FA1

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Question No:192

(Marks:1)

Vu-Topper RM

$a(a+b)^*b + b(a+b)^*a$ is the regular expression of language defined over $\Sigma=\{a,b\}$ that is _____.

- A. starting with b and ending in a
- B. starting with a and ending in a** ok
- C. starting with a and ending in b
- D. starting with a and ending in b or starting with b and ending in a

Question No:193

(Marks:1)

Vu-Topper RM

GTG for the expression $(a+b)^*bb$ may have minimum number of states:

Aaabcbbcbacc

Question No:194

(Marks:1)

Vu-Topper RM

Which of the following state is introduced while developing NFA for the closure of an FA?

- A. Final state
- B. Simply an initial state
- C. An initial state with loop for all letters
- D. An initial state which should be final as well** ok

Question No:195

(Marks:1)

Vu-Topper RM

In NFA, if null word (λ) is allowed to be a label of an edge, then that NFA is called _____.

- A. NFA with one string
- B. NFA with two strings
- C. NFA with null string** ok
- D. NFA without null string

Question No:196

(Marks:1)

Vu-Topper RM

Which one of the following is a correct word produced by the RE $(a^*b^*)ab$?

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A. Null

B. Abab ok

C. aaaa

D. bbbb

Question No:197

(Marks:1)

Vu-Topper RM

While developing NFA for the union of FA1 and FA2, if there is a loop of 'a' at the initial state of FA1 then the new initial state will have a transition for 'a' that goes straight to:

A. the final state of FA1

B. The initial state of FA1 ok

C. the initial state of FA2

D. the initial state of FA1*FA1

Question No:198

(Marks:1)

Vu-Topper RM

Let L be the language of all strings, defined over $\Sigma = \{0,1\}$, ending in 111. Which of the following strings are distinguishable with respect to L with z being 11?

111, 101

Question No:199

(Marks:1)

Vu-Topper RM

Which one of the following word is not accepted by the given regular expression?

abbbbaa

Question No:200

(Marks:1)

Vu-Topper RM

Which of the following is not a step-in elimination of states procedure?

Unify single transitions to multi transitions that contains union of input

Question No:201

(Marks:1)

Vu-Topper RM

In Moore machine the output depends on

The state

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Question No:202

(Marks:1)

Vu-Topper RM

While developing NFA for the union of FA1 and FA2, there will be

The initial state of FA1

Question No:203

(Marks:1)

Vu-Topper RM

Let FA3 be an FA corresponding to FA1FA2, then the final state of FA3 must correspond to the final state of

A. FA2 only **ok**

B. FA1 only

C. FA1 or FA2

D. FA1 and FA2

Question No:204

(Marks:1)

Vu-Topper RM

Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

A. FA1 only

B. FA2 only

C. FA1 or FA2

ok

D. FA1 and FA2

Question No:205

(Marks:1)

Vu-Topper RM

Mealy machine is equivalent to Moore machine, if we:

Applications of complementing and incrementing machines

Question No:206

(Marks:1)

Vu-Topper RM

In the context of make NFA for the concatenation of FA1 and FA2 (FA2 accepting null string), which of the following option is correct?

A. Final states in both FAs

ok

B. Initial states in both FAs

C. FA2 having final state only

D. FA2 having initial state only

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Question No:207

(Marks:1)

Vu-Topper RM

In the context of make NFA for the concatenation of FA1 and FA2 (none accepting null string), which of the following option is correct?

A. No final state in FA2 only

B. No initial state in FA1 only

C. No final and initial states in FA1 and FA2 respectively

D. No initial and final states in FA1 and FA2 respectively

Question No:208

(Marks:1)

Vu-Topper RM

Let FA1 accepts many strings and FA2 accepts no string, then $FA1+FA2$ will be equal to:

A. FA1

B. FA2

C. $(FA2)^*$

D. FA2-FA1

ok

Question No:209

(Marks:1)

Vu-Topper RM

The minimum length of the strings(except null string) of a language that starts and ends in the same letters will be:

A. 1

B. 2

ok

C. 3

D. 4

Question No:210

(Marks:1)

Vu-Topper RM

While developing NFA for the union of FA1 and FA2, there will be _____ transition/transitions for both 'a' and 'b' on the new initial state.

A. Single

B. Multiple

ok

C. Only one

D. Only three

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Question No:211

(Marks:1)

Vu-Topper RM

Which of the following form correctly expressed the regular expression RR^* ?

- A. R^+** **ok**
- B. R^-
- C. R^*
- D. $R+R^-$

Question No:212

(Marks:1)

Vu-Topper RM

Which of the following is not a step in elimination of states procedure?

- A. Unifying all the final states into one using e-transitions
- B. Get the resulting regular expression by direct calculation
- C. Remove states until there is only starting and accepting states
- D. Unify single transitions to multi transitions that contains union of input**

Question No:213

(Marks:1)

Vu-Topper RM

In the context of make NFA for the concatenation of FA1 and FA2 (Both FAs accepting null string), which of the following option is correct?

- A. Initial states in both FAs
- B. Final states in both FAs** **ok**
- C. FA2 having initial state only
- D. FA2 having final state only

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